

Magic Ball:

A globe-based system for multidimensional information display and interaction in MR

Lesi Hu, LSH, and Hu
School of New Media Art and Design,
Beihang University, China
1291620440@qq.com

Xiaozhan Liang, XZL, and
Liang
School of New Media Art and Design,
Beihang University, China
dearzhan1019@163.com

Shuo Yan, SY, and Yan*
School of New Media Art and Design,
Beihang University, China
yanshuobit@gmail.com

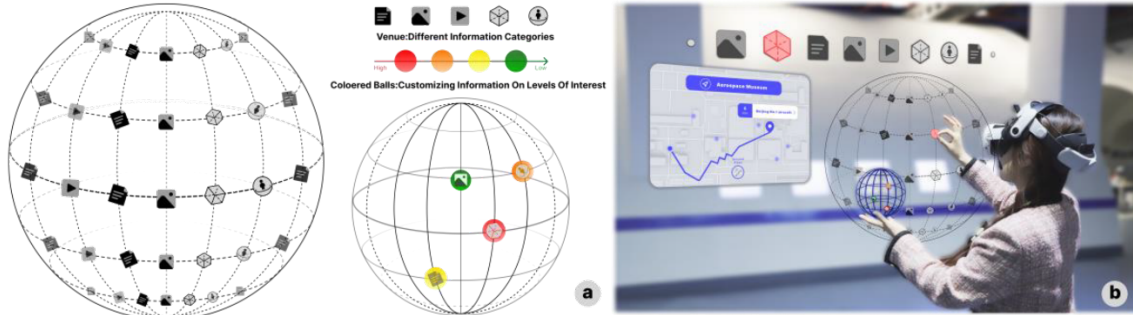


Figure 1: (a) Public ball and personal ball. (b) Complex information interaction system based on sphere

ABSTRACT

Recent advancements in Mixed Reality (MR) technology have catalyzed a profound reevaluation and transformation in the way information is presented in physical museums and exhibition spaces. MR technology transcends the limitations of traditional, flat presentations, integrating with museum and exhibition environments to enrich the levels of information display, thereby creating more captivating experiences for users. Current research primarily focuses on developing informational prompts and navigational aids within these spaces. However, there remains a notable deficiency in addressing users' deeper needs for spatial macro-cognition and interactive communication.

Our study delves into the advantages of globe-based information displays, merging them with the virtual reality aspects of MR technology. This integration forms a specialized 3D, multi-dimensional information display and spatial interaction system for museums and exhibition spaces. Utilizing the globe's longitudinal and latitudinal descriptions, the system exploits the unique attributes of the sphere. By differentiating personal and public spaces and integrating user behavior patterns with the characteristics of the globe, a range of globe-based interactive methods are designed, including activation,

picking up, throwing, navigating, and following. This culminates in a collaborative and shared experience for multiple users in a single setting, enhancing their spatial perception and interactivity.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Virtual reality**;

KEYWORDS

Information categorization, Globe, Spherical interface interaction, Multi-players exchange

ACM Reference Format:

Lesi Hu, LSH, and Hu, Xiaozhan Liang, XZL, and Liang, and Shuo Yan, SY, and Yan. 2024. Magic Ball:: A globe-based system for multidimensional information display and interaction in MR. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648668>

1 INTRODUCTION

Museums and exhibition spaces are quintessential venues of cultural and educational engagement. These settings, characterized by their complex informational landscapes, often pose challenges for users in attaining a comprehensive understanding of the information's nature and spatial arrangement.

The rapid emergence and evolution of Mixed Reality (MR) technology have markedly influenced these environments. Its integration into museums and exhibitions is becoming increasingly prevalent, offering enhancements in information presentation and spatial navigational aids. However, a significant gap remains in

*Corresponding author

addressing the macro-cognitive needs of users within these environments. This underscores the need for research focused on organizing information and developing display interaction strategies tailored to these complex settings.

This research is crucial, aiming to bridge the technological advancements with the macro-cognitive demands of users in these information-dense venues. The emphasis is on devising interaction strategies that not only ensure easy access to information but also augment user comprehension and engagement in these culturally and intellectually rich spaces.

Interface Presentation of Spherical. Recent research has classified immersive environments in terms of information categorized and displayed in 3D space [1, 2]. Meanwhile, a user-centered based curved interface is presented in a flat plane focusing on more content display [3], while it can reduce the user's burden. Therefore, the information display and visualization interfaces in immersive environments are more curved or shaped. Among them, globes have a long history as devices for displaying geographic information, with examples dating back to the Ancient Greece [4, 5]. Modern display technologies allow the use of globes with digital displays [6, 7], which studies have shown to have the universal cognitive interactivity and the possibility of unlimited data coverage, and for information more complex environments such as museums or exhibition spaces, globes have advantages for more multidimensional information displays.

Spatial Interaction Behavior of Globes: Explorations of globe-oriented interactions have been conducted in the new media [8–10], AR presentations [11], and professional data analysis [12], and has been used in many instances [13, 14]. In the context of immersive environments, [15–17] explored the use of a bigger visualization of light displays in a maritime museum based on a globe with a multipoint input approach, and also in terms of multiple harmony opportunities aspects of the globe; [18] performs knowledge of celestial motions by combining physical and virtual information about multiple planets, while also demonstrating the opportunities of the globe as a tool for presenting information in the field of education. However, currently, interactions for globes are dominated by tangible interactions, and such interfaces do not allow for real-time proximity to the user interface, and therefore, have spatial limitations. Therefore, in immersive environments, a virtual globe that is integrated with the real environment and accessible in real-time is valuable and currently available. virtual globes in immersive environments that are integrated with the real environment are of research value and are currently less explored.

In this paper, we designed a Magic Ball system to take the characteristics of globes with the virtual-reality fusion of immersive MR for 3D multidimensional information presentation and spatial interaction for use in a museum or exhibition space. The system includes a public ball and a personal ball (see Figure 1a). The ball uses the longitude and dimension attributes of globes to describe the type of information and spatial location respectively. We also expand the personal space and public space, exploring the application of interactive behaviors such as the wake of the personal menu, picking up, throwing, and navigating (see Figure 1b). Furthermore, our Magic Ball system shows a new way of collaborative communication between multiple people in the same scene.

Our Contribution as follows: Utilizing the user's general knowledge of the longitude and latitude of globes, we categorize and organize the information content of multi-dimensional information in complex information environments such as museums and exhibition spaces.

- Based on the globes, we have designed the interaction (rotating, pulling, picking, and throwing), and then constructed a real-time globes interaction system that can be followed;
- The system also takes advantage of the characteristics of the sphere to simultaneously construct a personal space and a public space, which can be interacted with in a multi-person environment to achieve information sharing and collection.
- Based on this, we deployed the system in the Aerospace Pavilion, using the system to organize the multiple categories and layers of information in the pavilion and to construct personal and public spherical spaces. For this, we conducted usability tests to test and verify the system.

2 DESIGN AND IMPLEMENTATION

2.1 Mapping of Latitude and Longitude on Globes

In complex environments like exhibition spaces, where information abounds, users often face challenges in developing a clear, macro-level understanding of the types and locations of information. The division of longitude and latitude forms the fundamental basis of our perception of globes. Globes, as devices for displaying geographical information, have a long history and are commonly found today in classrooms, libraries, and other educational settings, aligning with users' natural cognitive processes. Consequently, we leverage globes as a solution, mapping their longitude and latitude to categorize the types of content and spatial positions in complex information environments.

Longitude is Utilized to Map Different Types of Information. Considering the diversity of information in exhibition spaces, such as text, images, videos, models, and immersive visuals, we map various information types to distinct longitudinal axes. This allows users to selectively extract a single longitudinal ring to comprehensively view information of that type (see Figure 2a).

Latitude is Employed to Map Spatial Locations in The Real World. Reflecting the division of space into multiple floors in exhibition spaces, such as the first, second, and third floors, we map information from different floors to corresponding latitudinal axes.

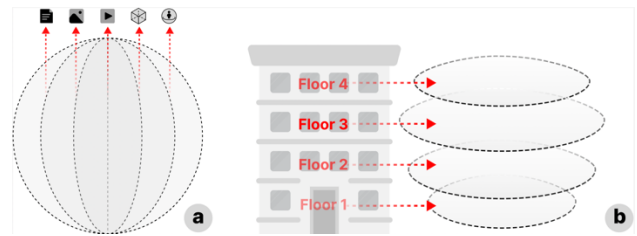


Figure 2: (a) Longitude: Classification of information type. (b) Latitude: Geographical location of the real space

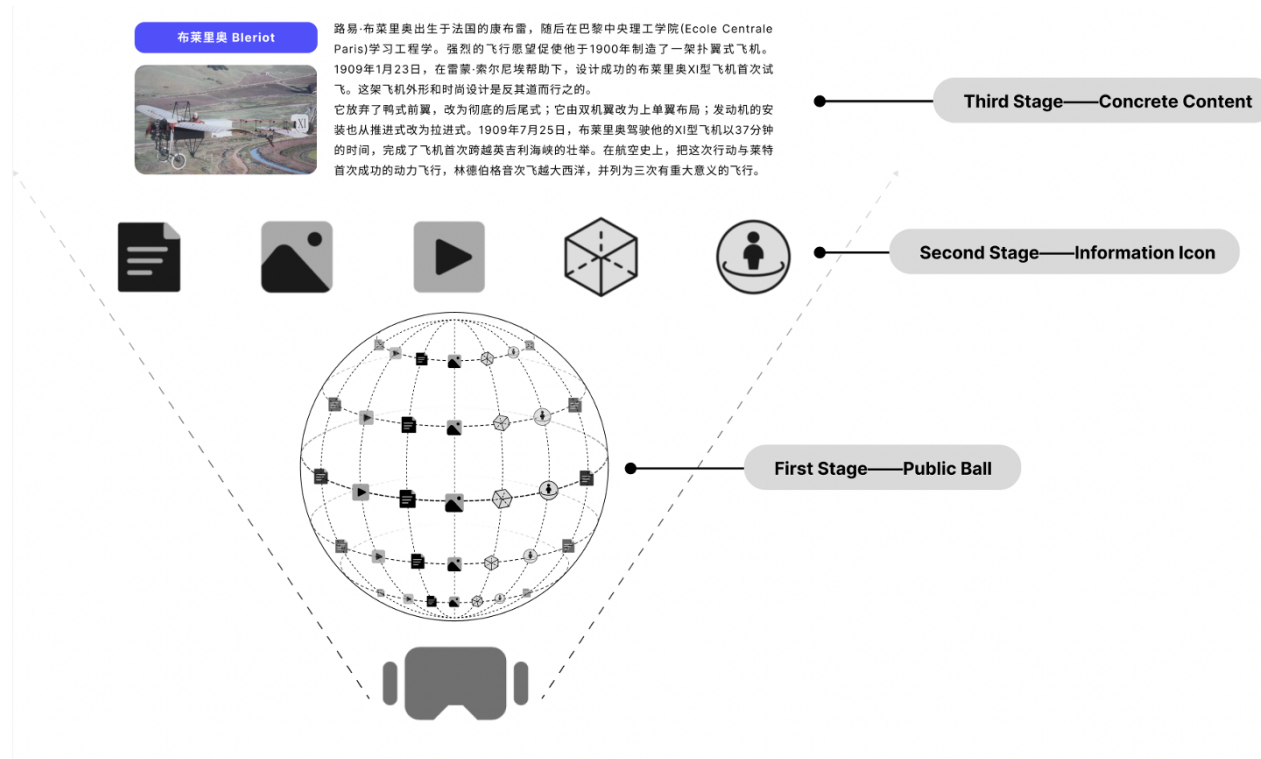


Figure 3: Information structure of the Magic Ball

Users can thus extract a single latitudinal ring to holistically view all information on that particular floor (see Figure 2b).

2.2 Interaction based on globe features

The overall information architecture comprises three levels. The first level is the public ball, which, upon user interaction, triggers the second-level information icons. A further click on these icons then brings up the third level of specific information (see Figure 3).

Utilize Intersections for Specific Data Insights: In globes, the separate mapping of latitude and longitude creates numerous intersecting points. Each point, embedded with a mini globe, signifies a distinct mini point. Users can interact with a mini-point by clicking on its globe to access detailed information (see Figure 4a). Latitude symbolizes real-world spatial relationships, like the floors in a mall or the arrangement of exhibits in a museum, with different latitudes denoting these real-space positions (see Figure 4b). Longitude represents various types of data, categorized into text, images, videos, models, and immersive footage, with different meridians indicating these distinct data types (see Figure 4c).

Globe-Based Interface Control through Rotation. Miniature interaction is an intuitive method. Leveraging the natural human interaction with spheres—rotation—we map the rotation degree to the scrolling range of the interface. When users view information along a specific longitude or latitude and the expanded content becomes too extensive, partly hiding the information and prompting users to scroll, a globe exists between the user and the

expanded interface, at a relatively far distance. In this context, manipulating the miniature globe in front of them becomes a natural and intuitive way of interaction. Users can select a level and rotate the globe left or right to scroll the interface left or right, respectively, to reveal more information (see Figure 4d).

Immersive Browsing within Globes. The closed space formed by the spherical curved surface, commonly seen in astronomical exhibits, offers a strong sense of immersion. In our system, we've developed an immersive browsing feature for three-dimensional panoramic data. Users can enter the interior of globes for an immersive perspective to browse information (see Figure 4e).

2.3 Globes for Personal and Public

To differentiate personal and public spaces in MR, we've focused on the uniqueness of personal spaces and the interactivity in multi-user scenarios to develop the 'personal ball' and 'public ball' (see Figure 5a). These offer users a more enriched and personalized MR experience.

Palm Wake and Real-Time Following Design. Our design incorporates a mechanism of palm wake and real-time following. In the MR environment, the public ball constantly follows in front of the user. In contrast, the personal ball is activated by the user's palm-opening action, catering to their individual needs (see Figure 5b).

Presenting Information in Personal and Public Spaces. The public ball contains general information about the exhibit

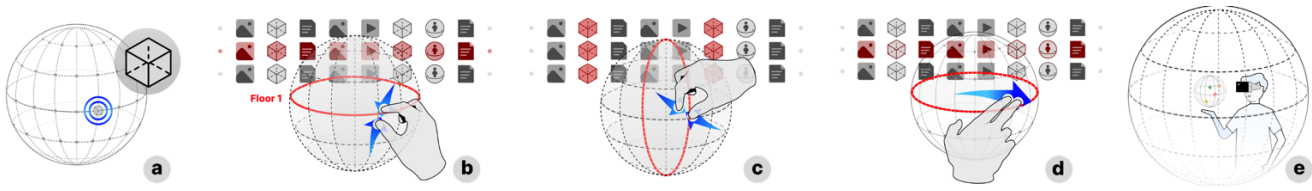


Figure 4: (a) Click: Select the mini point to view information. (b) Pull-out: view the single layer of information. (c) Pull-out: View category information. (d) Slide: Rotate the public ball to control the interface scrolling. (e) In the Globe: View panoramic information.

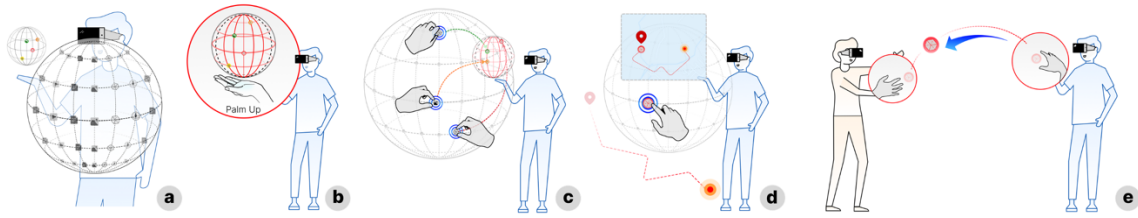


Figure 5: (a) Relationship of the public and the personal ball. (b) Palm up to wake up the personal ball. (c) Grab the mini point to the personal ball. (d) Navigation: follow the ball to the location. (e) Multiplayer exchange: collaborative sharing of mini point.

and spatial relationships, representing public content. Simultaneously, users can customize their personal space—the personal ball—according to their preferences and needs. They can select mini-points of interest and drag them into their personal ball for collection. These mini-points automatically adhere to the corresponding intersecting points of latitude and longitude within the personal ball based on their category and location (see Figure 5c). Moreover, during this process, users can choose colors for the selected mini-points to facilitate intuitive differentiation of information in the personal ball. Here, four colors from green to red represent increasing levels of interest.

Spatial Navigation of Information on the Ball. Users can select a mini-point on the ball and say, ‘Take me there.’ Following voice recognition, a route map to the destination will appear in front of the user (see Figure 5d). They can then navigate following the map’s route until they reach the destination.

Pitch Interaction and Communication Sharing. In multi-user scenarios, users can ‘pitch’ information to share it with others in the same space (see Figure 5e), thus facilitating information exchange and sharing.

3 CONCLUSION AND FUTURE WORK

In summation, while the Magic Ball system demonstrates proficiency in usability and learnability in the user experiment. It presents avenues for further enhancement. Future iterations should focus on streamlining the operational process and incorporating more intuitive learning mechanisms to elevate the overall user experience. By combining multiple data visualization techniques, we can more fully understand user needs and guide continuous improvement of the system.

REFERENCES

- [1] Benjamin Lee, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2022. A Design Space For Data Visualisation Transformations Between 2D And 3D In Mixed-Reality Environments. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 25, 1–14. <https://doi.org/10.1145/3491102.3501859>
- [2] C. Plasson, R. Blanch and L. Nigay, “Selection Techniques for 3D Extended Desktop Workstation with AR HMD,” 2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Singapore, Singapore, 2022, pp. 460-469, doi: 10.1109/ISMAR55827.2022.00062.
- [3] Jiazhou Liu, Barrett Ens, Arnaud Prouzeau, Jim Smiley, Isobel Kara Nixon, Sarah Goodwin, and Tim Dwyer. 2023. DataDancing: An Exploration of the Design Space For Visualisation View Management for 3D Surfaces and Spaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 379, 1–17. <https://doi.org/10.1145/3544548.3580827>
- [4] Edward Luther Stevenson. 1921. Terrestrial and celestial globes: Their history and construction, including a consideration of their value as aids in the study of geography and astronomy. <https://www.gutenberg.org/files/39866/39866-h/39866-h.html>
- [5] Wikipedia contributors. 2021. Globe. <https://en.wikipedia.org/wiki/Globe>
- [6] Hrvoje Benko, Andrew D Wilson, and Ravin Balakrishnan. 2008. Sphere: multi-touch interactions on a spherical display. In *Proceedings of the 21st Annual ACM Symposium on User Interface Software and Technology*. ACM, Monterey, CA, USA, 77–86. <https://doi.org/10.1145/1449715.1449729>
- [7] Nikita Soni, Schuyler Gleaves, Hannah Nef, Sarah Morrison-Smith, Shaghayegh Esmaeili, Ian Mayne, Sayli Bapat, Carrie Schuman, Kathryn A Stofor, and Lisa Anthony. 2020. Adults’ and Children’s Mental Models for Gestural Interactions with Interactive Spherical Displays. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama, Japan, 1–12. <https://doi.org/10.1145/3313831.3376468>
- [8] Seán Clarke, Antonio Voce, Pablo Gutiérrez, and Frank HulleyJones. 2020. How coronavirus spread across the globe –visualised. <https://www.theguardian.com/world/ng-interactive/2020/apr/09/howcoronavirus-spread-across-the-globe-visualised>
- [9] Josh Dye, Craig Butt, Richard Lama, and Mark Stehle. 2020. Silent Skies. <https://www.theage.com.au/interactive/2020/coronavirus/silent-skies/index.html>
- [10] Raluca Nicola. 2018. World population count. https://ralucanicola.github.io/JS-API_demos/world-population/
- [11] Marlon Lückert. 2020. CovidAR - Data Visualization in Augmented Reality. <https://devpost.com/software/covidar-data-visualization-in-augmentedreality>
- [12] Mark Billinghurst, Hirokazu Kato, Ivan Poupyrev, et al. 2008. Tangible augmented reality. *ACM SIGGRAPH Asia* 7, 2 (2008), 1–10.

- [13] Tom Chandler, Maxime Cordeil, Tobias Czauderna, Tim Dwyer, Jaroslaw Glowacki, Cagatay Goncu, Matthias Klapperstueck, Karsten Klein, Kim Marriott, Falk Schreiber, *et al.* 2015. Immersive Analytics. In 2015 Big Data Visual Analytics (BDVA). IEEE, Hobart, Tasmania, Australia, 1–8. <https://doi.org/10.1109/BDVA.2015.7314296>
- [14] Kim Marriott, Falk Schreiber, Tim Dwyer, Karsten Klein, Nathalie Henry Riche, Takayuki Itoh, Wolfgang Stuerzlinger, and Bruce H Thomas. 2018. Immersive Analytics. Vol. 11190. Springer, Berlin, Germany.
- [15] Nikita Soni. 2020. Designing Interactions for Multi-touch Spherical Displays to Support Collaborative Learning in Museums. In Companion Proceedings of the 2020 Conference on Interactive Surfaces and Spaces (ISS '20). Association for Computing Machinery, New York, NY, USA, 95–99. <https://doi.org/10.1145/3380867.3427408>
- [16] David Englmeier, Wanja Sajko, and Andreas Butz. 2021. Spherical World in Miniature: Exploring the Tiny Planets Metaphor for Discrete Locomotion in Virtual Reality. In 2021 IEEE Virtual Reality and 3D User Interfaces (VR). IEEE, Lisboa, Portugal, 345–352. <https://doi.org/10.1109/VR50410.2021.00057>
- [17] David Englmeier, Isabel Schönewald, Andreas Butz, and Tobias Höllerer. 2019. Feel the globe: Enhancing the perception of immersive spherical visualizations with tangible proxies. In 2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE, Osaka, Japan, 1693–1698. <https://doi.org/10.1109/VR.2019.8797869>
- [18] Yusuke Morita and Norio Setozaki. 2017. Learning by tangible learning system in science class. In International Conference on Human-Computer Interaction. Springer, Vancouver, Canada, 341–352. https://doi.org/10.1007/978-3-319-58077-7_27